Daily Drilling Report
SPD19B-03
9/3/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
38.0	12 1/4"	183.00	72.83	31.70	104.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	220,877.90		9,090,327.02		33.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,725.00	2,725.00	0.00			

Operations at Report Time
L/D old 12 1/4'' section BHA.

Operations Summary
Cont' RIH 9 5/8" - 10 3/4'' CSG f/ 1165m to 2720.8m (Shoe Depth). Land CSG HGR. Perform CMT job. Displace cement and bump plugs. Perform CSG pressure test, OK.

Operations next 24hrs
L/D cement head. B/O HGR RT. R/D CSG handling tools. Wash inside WH. Set seal pack off. L/D old 12 1/4" BHA. Perform BOP test. Install short wear bushing. RIH 8 1/2" directional BHA#10. D/O 9 5/8" shoe track. Perform FIT. Drilling 8 1/2" hole.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah / Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/31/2014	10/1/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 31

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	02:30	2.50	12 1/4"	G	18	P	Cont' RIH 9 5/8" CSG (47 PPF, L-80, TMK) f/ 1165m to 1487m (13 3/8" CSG shoe).
02:30	03:30	1.00	12 1/4"	G	07	P	Engage Fill up tool and circulate 430 bbl (1.2 string volume) w/ 450 gpm/ 420 psi.
03:30	12:00	8.50	12 1/4"	G	18	P	Cont' RIH 9 5/8" CSG (47 PPF, L-80, TMK) f/ 1487m to 2477m. * Break circulation every 20 jts while RIH. * Break circulation at 2200 m, observed shoe plugged. Work on string by pressure and jercking. Plug free by 800 psi. * Install Bow Centralizers around Stop Collar as per tally.
12:00	13:00	1.00	12 1/4"	G	18	P	M/U X/O Jt (9 5/8" TMK pin x 9 5/8" Vam Top box). Change out 9 5/8" CSG handling tools to 10 3/4". M/U X/O (9 5/8" Vam Top pin x 10 3/4'' Tenaris Blue box) + 1 joint 10 3/4'' casing (55.5 ppf, L-80, Tenaris Blue) in one piece and torque up the connection as per the thread type. RIH to 2501m.
13:00	15:30	2.50	12 1/4"	G	18	P	Cont' RIH 10 3/4" CSG (55.5 PPF, L-80, Tenaris Blue) f/ 2501m to 2698m.
15:30	16:00	0.50	12 1/4"	G	18	P	P/U & M/U 10 3/4" Aker casing hanger Assy: X/O (10 3/4" VamTop box x 10 3/4" Tenaris Blue Pin) + Nipple (10 3/4" VamTop box x 10 3/4" VamTop Pin) + Aker casing hanger + RT + 3m pup jt + 5m pup jt
16:00	17:00	1.00	12 1/4"	G	18	N	Sting in fill up tool. Try to wash down, obsereved shoe plugged. Work on string by moving string up and jerking down while keeping 1000 psi pressure in string, no success. Bleed off pressure. increase pressure in string up to 2000 psi and continue working on string, observed pressure plummeted, shoe unplugged and established normal circulation.
17:00	17:30	0.50	12 1/4"	G	18	P	P/U landing joint & Wash down to 2720.8m w/ 300 gpm, 1000 psi and land casing hanger inside Unihead. * Float shoe @ 2720.8m , Float collar @ 2696.5m. * Totally run 2 jts 9 5/8" casing (47 ppf, L-80, Vam Top) + 1 X/O joint (9 5/8" Vam Top pin x 9 5/8" TMK box) + 207 jts 9 5/8" casing (47 ppf, L-80, TMK) + 1 X/O (9 5/8" TMK pin x 9 5/8" Vam Top box) + 1 X/O (9 5/8" Vam Top pin x 10 3/4" Tenaris Blue box) + 18 jts 10 3/4" casing (55.5 ppf, L-80, Tenaris Blue) + X/O (10 3/4" VamTop box x 10 3/4" Tenaris Blue Pin) + 1 no Nipple (10 3/4" VamTop box x 10 3/4" VamTop Pin) + 10 3/4" Hanger + 128 no 9 5/8" x 12 1/4" Botil bow type centralizer & stop collar.
17:30	18:00	0.50	12 1/4"	H	01	P	P/U & M/U Mehran X/O and N/U Mehran Double CMT Head (Top and Btm CMT Plugs already loaded). R/U surface line.
18:00	20:30	2.50	12 1/4"	H	07	P	Circulate 1100 bbl w/ 450 gpm/ 1300-1050 psi.

Daily Drilling Report
SPD19B-03
9/3/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
20:30	22:30	2.00	12 1/4"	H	09	P	HPJSM. L/U to CMT unit , flush and pressure test surface lines to 500/4500 psi for 5/10 min, Ok. Pump 40 bbls Chemical wash + 70 bbls of 12.4 ppg Spacer w/ 8 bpm/ 900 psi. Drop Btm CMT Plug. Pump 498 bbl 12.7 ppg Lead cement slurry (Lead Solution = 387 bbl) w/ 8 bpm / 1000-650 psi. Cont' pump 59 bbls of 15.8 ppg Tail CMT slurry w/ 6.5 bpm/ 500-400 psi, (Tail Solution = 35 bbl). Drop Top CMT Plug, displace CMT slurry w/ 10 bbls of 12.4 ppg Spacer + 10 bbls of 12.1 ppg mud w/ 8 bpm, 450 psi. * Totally used 87 MT of Cement "G". * Tail slurry Chemical usage : Anti foam: 27 gal, Fluid loss: 50 lbs, Dispersant: 303 lb, Retarder: 101 lbs, Gas block: 806 gal, Stabilizer: 81 gal. * Lead Chemical usage : Bent: 3655 lbs, Retarder: 1023 lbs, Antifoam: 62 gal, Sod. Silicate: 187 gal. * Chemical wash usage: Chemical wash: 20 gal, Solvent: 10 gal * Spacer usage: Spacer: 850 lbs, Anti foam: 5 gal, Barite: 19850 lbs.
22:30	00:00	1.50	12 1/4"	H	07	P	Switch to rig pump and cont' displacing CMT slurry by 642 bbls of 12.1 ppg mud (totally 662 bbls) w/ 450-80 gpm, 600-400 psi. CMT Plug not bumped. Cont' displacing 2 bbls more, plug bumped and increase pressure to 2500 psi. Switch to CMT unit, increase pressure to 4000 psi/10min for casing pressure test, ok. Final differential press 400 psi w/ 80 gpm. Check float holding, ok. * No loss during cementing and displacing. * Dump 40 bbls Chemical wash + 70 bbls Spacer + 30 bbls contaminated CMT + 20 bbls Neat CMT.
00:00	00:00						Sep 04 , 2014 00:00 - 00:30 B/O surface lines and R/D Mehran CMT Head & X/O. Meanwhile wash inside WH through side outlet valves by CMT unit. 00:30 - 01:00 B/O & L/D landing jt, CSG hanger R/T + 2 x Pup Jts as one piece. 01:00 - 02:30 R/D Al-Ghaith equipment. R/U 5 7/8" handling equipment. 02:30 - 03:30 P/U & M/U AKER 13 5/8" washing tool to one std 5 7/8" DP, RIH and wash inside BOP and Unihead and top of 10 3/4" CSG HGR w/ 600 gpm , 5 rpm, 60 psi pressure. Meanwhile wash inside WH through side outlet valves by CMT unit. POOH and L/D washing tool. 03:30 - 05:00 P/U & M/U 10 3/4" Aker Pack-off Assy + R/T+ one std of 5 7/8" DP. RIH @ 18.8m, L/U CMT unit to Choke line, flush surface lines and pressure test same to 500/5000 psi for 5/10 min, OK. Close Upper Ram. Set Pack-off Assy by apply 2000 psi pressure in annulus. Increase pressure to 5000 psi for energizing, OK. Pull with 10 Kips over pull and release RT. POOH to surface, B/O & L/D R/T. Perform pressure test elastomeric seal area from Uni-Head port by hand pump to 5000 psi/10 min, OK. 05:00 - 06:00 L/D old 12 1/4" section BHA consists of: 9 5/8" Motor + 8" PNMDC + 11 7/8" String Stabilizer + 1 jt 8" NMDC.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,725.00	KCl-Polymer Mud	12.10				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
5278.0						

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.10	6.	359	1.0	998.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mataf Pride	Crew Boat	9/3/2014 20:00	9/3/2014 21:00	Disemb 11 PAX and Embarked 09 PAX for Kish. C/Off and departed for Kish.
Swissco	Supply Boat	9/3/2014 14:00	9/3/2014 15:00	B/Loaded Cargo for Aban 8, C/Off & Dep for Aban 8 via Sahar 2.
Forte	Supply Boat	9/3/2014 12:00		Delivered FW-120 MT + DD6 Cargo & B/Loading Cargo.

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB		159.0	57.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Cement G	87.0	0.0	92.0
Drill Water	0.0	120.0	343.0

Daily Drilling Report
SPD19B-03
9/3/2014

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Diesel Fuel	7.0	0.0	167.0
Potable Water	23.0	15.0	85.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")

TFA (in²)	Drilling Time (hrs)	Bit Revs	Depth In (mKB)	Depth Out (mKB)	Depth Drilled (m)	BHA ROP (m/hr)	IADC Bit Dull

Flow Rate (min) (gpm)	Flow Rate (max) (gpm)	SPP (min) (psi)	SPP (max) (psi)	WOB (min) (1000lbf)	WOB (max) (1000lbf)	RPM (min) (rpm)	RPM (max) (rpm)

Circulating Time (hrs)	Rotating Time (hrs)	% Rotating Time (%)	% Sliding Time (%)	PU HL (max) (1000lbf)	Rot HL (max) (1000lbf)	SO HL (min) (1000lbf)	String Wt (1000lbf)

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.4

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
1,200.0	7.7	1,200.00		14.03	3,413.7

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S4		2,594.00	2,457.78
S3		2,622.00	2,483.90
S2		2,632.50	2,493.72
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	

General Remarks

Comment

>> Requirements: (MI MR # 14 & 15)

>>> Sent:

** Swissco to ABAN-8: (15 BB CaCO3 - F)

>>> Received:

** FORTE f/ Kish: (57 jts 7" Liner, 29 ppf, L-80, Vam Top) + (27 jts 7" TBG, 26 ppf, 28 % CR, 110 KSI, Vam Top) + (2 no 7" Float Shoe) + (2 no 7" Float Collar) + (3 no 7" Landing Collar) + (5 no Thread Compound) + (3 no 7" Pup Joint, 3m Long, 29 ppf, L-80, 13 % CR) + (1 no 7" Pup Joint, 4m Long, 26 ppf, K-110) + (6 no 7" Wiper Plug) + (4 no 5 7/8" Pump down plug) + (1 no 5 1/2" Pump down plug) + (100 no 7" Spirolizer) + (200 no Spir-o-lok Stop Collar) + (1 no 7" Drift, 26 ppf) + (120 MT DW).